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SIT233 – Cloud Computing
Module 11 Guided Lab - Streaming Dynamic Content using Amazon CloudFront

Task Questions

1. Question 1

I can set up many origin kinds for CloudFront to serve my content. These include file storage options like Amazon S3 buckets, HTTP/HTTPS servers such as EC2 instances or any other type of web server, live video streaming options like Amazon MediaPackage and low-latency live media streams like Amazon MediaStore, and live video transit options like AWS Elemental MediaConnect.

2. Question 2

I put up an Origin Access Identity (OAI) or Origin Access Control (OAC) to make sure my S3 bucket only accepts requests from CloudFront. Initially, I open the CloudFront dashboard and create an OAC. I then amend my S3 bucket policy to allow the OAC to access it. This entails adding a particular policy that permits CloudFront to access the bucket and modifying the bucket policy under the S3 console's Permissions tab. Lastly, I verify that the configuration is working by trying to visit the S3 bucket directly, which returns an access denied error, but using the CloudFront URL, which allows access. My S3 bucket is safe and only accessible through CloudFront thanks to this configuration.

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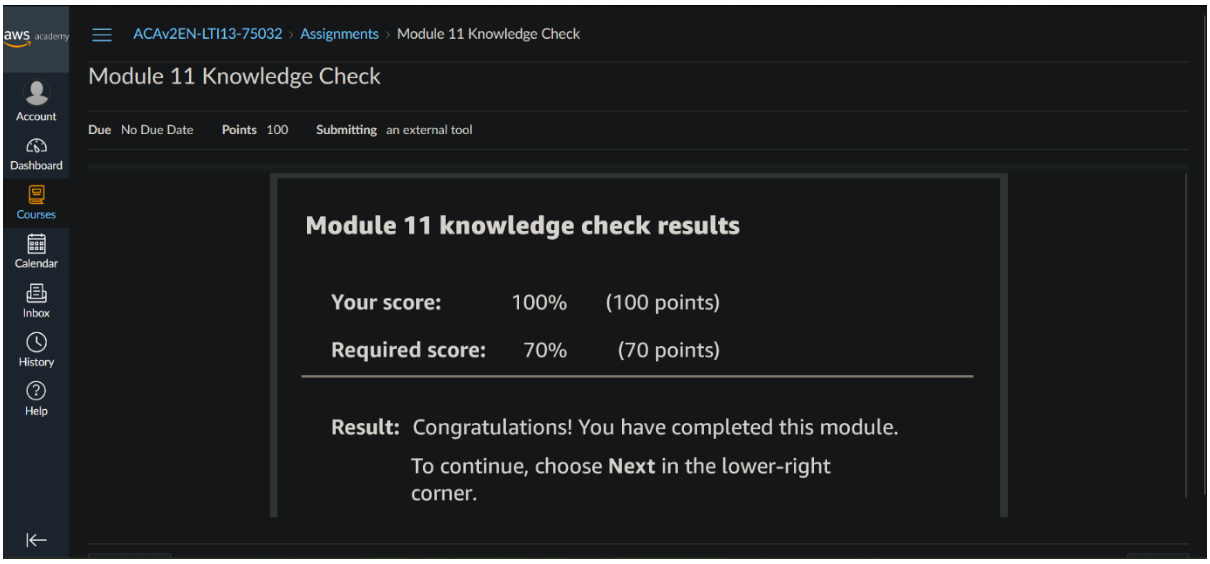
Module 11 Knowledge check

This knowledge check was comprised of the most important questions throughout the module 11. The questions asked was kind of a summary to the whole module 11. In this knowledge check I was able to see my understanding of this module, and the from the mistakes I was able to correct myself of the parts that I was not sure about.

This knowledge check was mainly about Caching. First question itself was about Caching which is a high-speed data storage layer. Then there were questions about types of cache and benefits of cache. Then how cloudfront uses caching. Where application session data are cached. Best practices of caching, how DynamoDB uses caching and about elasticache.



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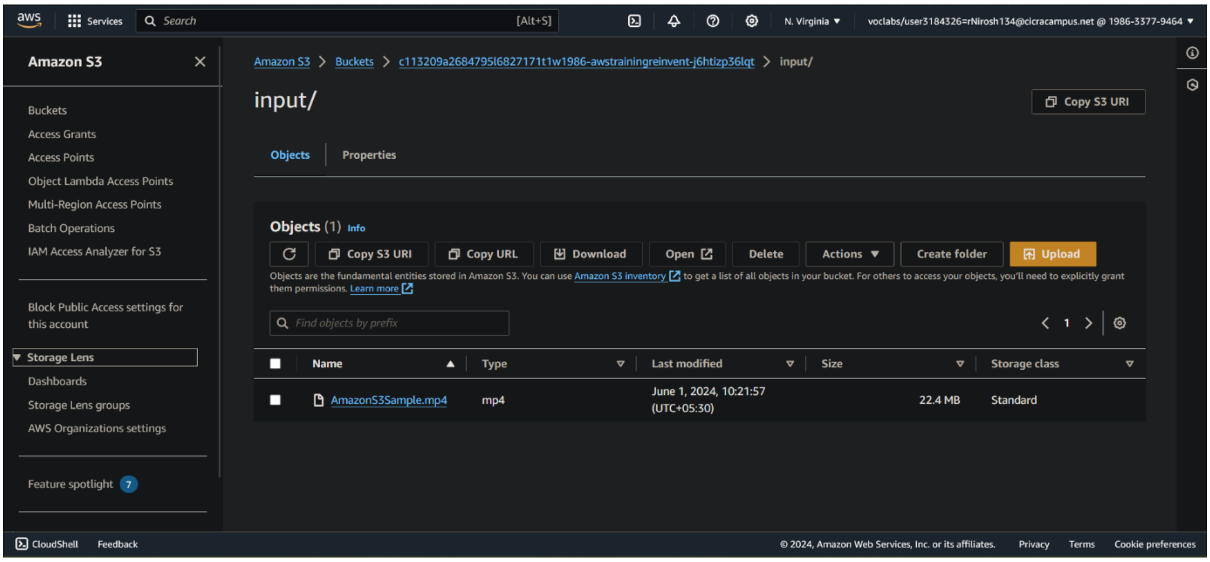
Task 1: Lab Preparation

I set up an Amazon S3 account during the exercise session in order to get ready to create a video stream that adjusts to various internet speeds. I made advantage of an internet S3 storage pre-loaded video file.

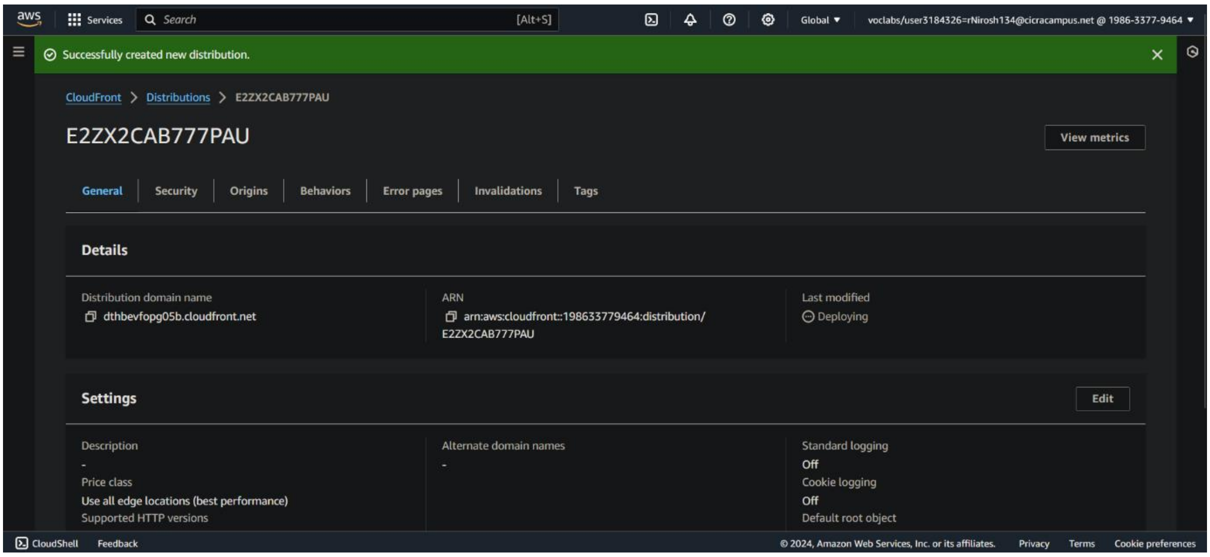
Initially, I used the Amazon S3 console to log in and searched for the "awstrainingreinvent" storage container, which had the practice video. I also took note of this container's location. I discovered a "input" folder inside, including the "AmazonS3Sample.mp4" video clip.



Task 2: Create an Amazon CloudFront Distribution

To deliver the video at various quality levels, I created a solution on Amazon called CloudFront in this stage of development. In this manner, the video would still be accessible anyone with lesser internet rates.

I started by developing a new distribution and moving to the CloudFront service. Next, I selected the "awstrainingreinvent" S3 bucket, which is the same one I used previously, to tell CloudFront where to receive the video files. I didn't apply any additional security measures and instead left the files set to public access for this practice session. After finalizing all the details, I generated the CloudFront distribution.



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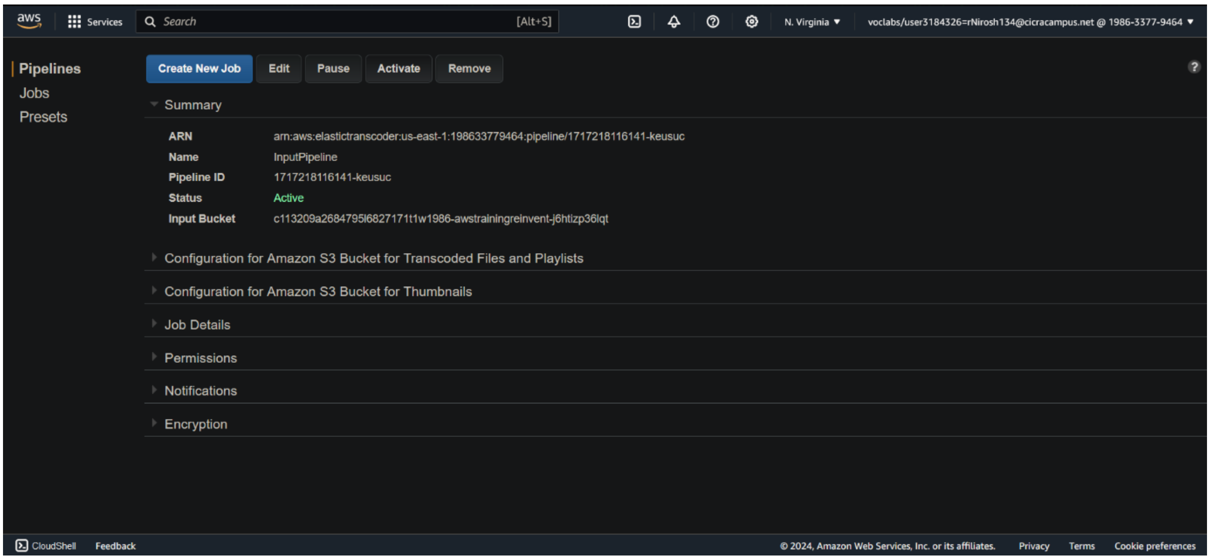
Task 3: Create an Amazon Elastic Transcoder Pipeline

In this step, I set up an assembly line on Amazon called Elastic Transcoder to process the video.

This assembly line, called a pipeline, would handle jobs that convert the video file into different qualities.

First, I switched to the Elastic Transcoder service and made sure the region matched where our S3 bucket was located. Then, I created a new pipeline named InputPipeline. For this practice session, I told the pipeline to use the same S3 bucket, awstrainingreinvent, to store both the original and converted video files. I also chose a pre-loaded security role to allow Elastic Transcoder to access the bucket.

I then made a job in the pipeline. The video file would be taken on this task and transformed into several variants at various bit-rates. I titled the job and instructed it to search the input folder of the S3 bucket for the video file "AmazonS3Sample.mp4".

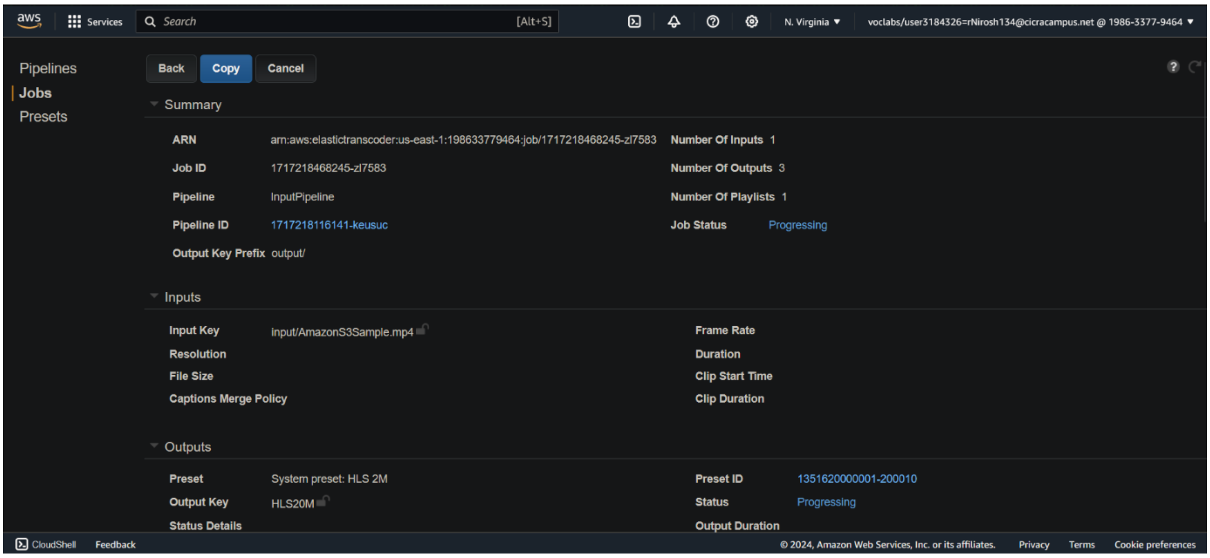


Then I set up the job to produce three separate video versions, each with a different bit-rate appropriate for varied internet speeds. I gave each bit-rate a distinct name prefix and set up the video to be split into 10-second chunks.

Making a "playlist" that would serve as a menu for viewers was the final stage. The three different bit-rates of the video would be combined into a single source using this playlist. The gadget used by viewers could then select the version that maximizes internet speed.

I named the playlist and instructed it to contain the three previous video versions that I had made.

When everything was finally in place, I approved the creation of the job. The actual conversion procedure should to take no more than a minute.



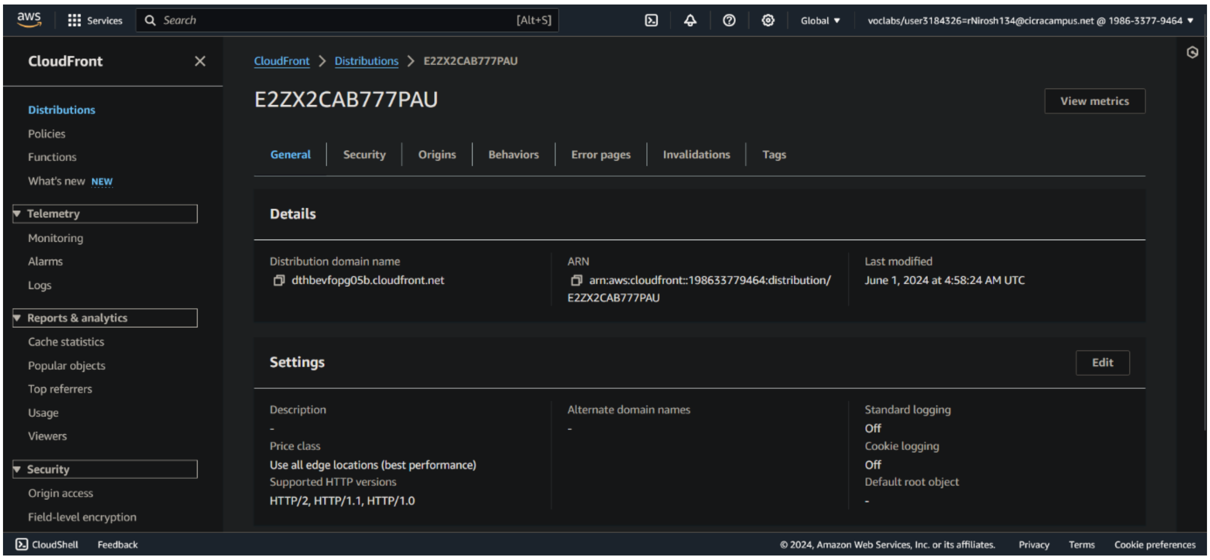
Task 4: Test Playback of the Dynamic (Multi Bit-Rate) Stream

In the final stage, I verified that the various quality video streams were functioning. To test this, I used my phone.

should utilize the default web browser on my phone because some web browsers might not operate.

I created a unique web address (URL) that led to the video stream in order to test it. This URL was composed of two parts: the location of a file (playlist) that contained a list of the video attributes, and the web address from a service called CloudFront, which looked like a website address.

By examining the object I had previously built I was able to locate the CloudFront address in the CloudFront service. After setting it up, I copied the URL.



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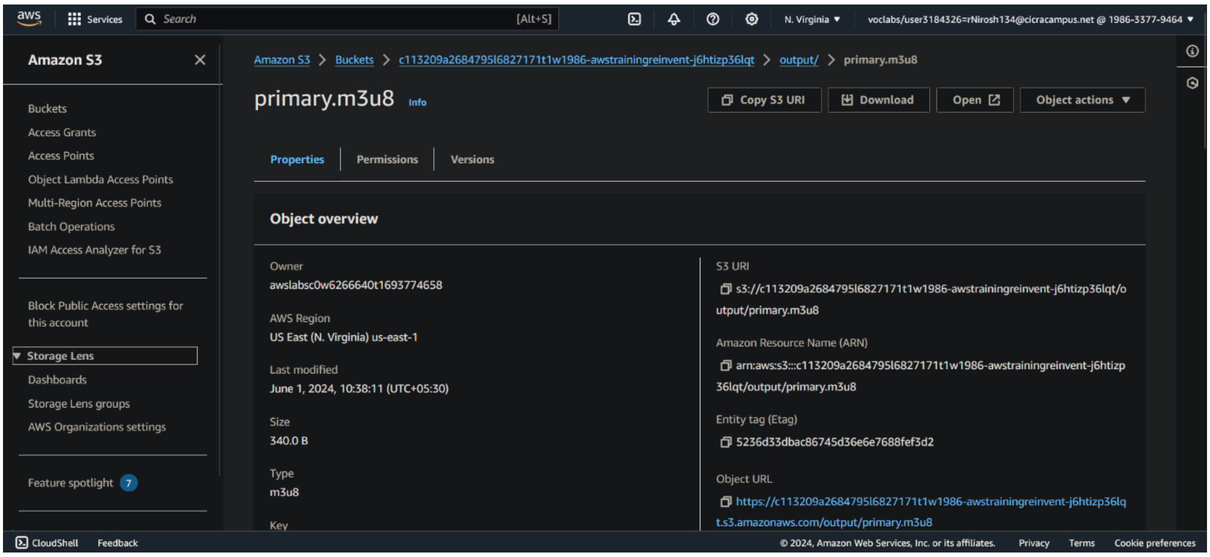
Upon returning to the storage service (S3), I discovered a file called "primary.m3u8" within the "output" folder (which included the video files that had been converted). The playlist was contained in this file.



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Finally, I appended "/output/primary.m3u8" to the end of the CloudFront web address. I entered the entire URL into the web browser on my phone.

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Grade:

My Grade is 40/40